

Hearing the regularity of a diffusion coefficient: a graded local Weyl law with a constructive converse

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Abstract

The on-diagonal heat kernel of $L = -\partial_x(c(x)\partial_x)$ carries a local Weyl remainder $r_t(x) = K_t(x, x)\sqrt{4\pi c(x)t} - 1$. This is a paper of *experimental mathematics with precise conjectures*: we present a window heuristic and systematic fixed-seed evidence — not proofs — that the decay rate of $\sup_x |r_t|$ *grades* the regularity of the coefficient (Conjecture G: C^α cusps read as $t^{\min(1, \alpha/2)}$, coefficients in the Favard/saturation class $\{c : c'' \in L^\infty\}$ as t^1), and we study the converse: does a full-rate reading force membership in the regularity class? For the *practically available* (coefficient-free) form of the detector, as a statement over any *finite window of scales* the converse is **false**, by homogenization: a coefficient with fine (analytic, but large-norm) microstructure reads, at every scale above its cell size, as a smooth *effective* medium, the fitted coefficient agreeing with the classical harmonic mean to 0.6%. Every such impostor carries its own expiry scale, breaking away at $O(1)$ below the crossover $t \sim \delta^2$. The repaired statement is therefore scale-quantified — full-rate decay *at all scales* conjecturally characterizes the class (Conjecture C below; the frame is the semigroup characterization of Hölder–Besov spaces, the proof obligation — uniform two-sided parametrix bounds for merely-Hölder coefficients — is stated and not discharged) — and it is probed constructively: from detector data alone we recover the Hölder exponent to ± 0.03 (and, in a de-circularized variant that never consults the true coefficient, to comparable accuracy), locate the singularity to 7×10^{-4} (twenty times finer than the probing kernel’s width), verify a two-sided near-linearity of the signal in the cusp amplitude (mildly sublinear: signal/amplitude drifts from 0.073 to 0.065 over an $8\times$ range — the honest spread $\times 1.12$), and classify smooth / cusp / scale-break cases blind at 12/12, with the decision thresholds dictated by the graded law itself (admissible exponents lie in $(0, 1]$, so a non-decaying reading within the window belongs to no Hölder class at those scales and is an impostor signature). An adversarial study shows a genuine cusp cannot be hidden by smooth-plus-oscillatory dressing across the scale ladder. A load-bearing numerical lemma is included: the lattice artifact of the discrete kernel admits an exact constant-coefficient rescaling baseline, $K_t(c) = K_{ct}(1)$, whose subtraction improves the detector floor from 10^{-2} to 10^{-7} — without it, every measurement above is noise. Scope: one dimension; the multidimensional and Lorentzian ports are discussed.

1. Introduction

Weyl asymptotics and heat-kernel expansions are the classical bridge between spectrum and geometry [1, 2, 3]. For smooth coefficients the on-diagonal kernel of $L = -\partial_x(c \partial_x)$ on the circle satisfies $K_t(x, x) = (4\pi c(x)t)^{-1/2}(1 + a_1(x)t + \dots)$ with a_1 built from c'' , $(c')^2/c$; the *local Weyl remainder*

$$r_t(x) = K_t(x, x)\sqrt{4\pi c(x)t} - 1$$

then decays linearly in t . When c is merely Hölder the expansion fails, and the natural question — in the spirit of "can one hear the shape of a drum" [4] but local and quantitative — is what the decay of r_t *hears*.

Conjecture G (the graded law, forward direction) — and its epistemic status. For c with a C^α cusp, $\sup_x |r_t| \sim t^{\min(1, \alpha/2)}$. The heuristic: the kernel averages the coefficient over windows of width $\sqrt{t}$, so the remainder is a Gaussian-window modulus of continuity at scale $\sqrt{t}$. We are explicit that this paper does **not** prove the law — that would require uniform two-sided parametrix bounds for merely-Hölder coefficients, which we do not supply; what we supply is the heuristic, sharp receipts (measured rates 0.232/0.494/0.969 against 0.25/0.50/1.00 for $\alpha = 0.5, 1.0$ and smooth; the same rates at $N = 768$ agree to within 0.001 — Section 6), and the two-sided amplitude evidence of Section 4. We also identify why the band $\alpha \in (1, 2)$ is harder to resolve numerically (Section 6). Here and throughout, "smooth" as a detector verdict means the **saturation class**: the $a_1(x)t$ term caps the readable rate at t^1 , and by classical saturation theory for first-order semigroup approximation the $O(t)$ class is the **Favard class** $\{c : c'' \in L^\infty\}$ ($C^{1,1}$ -type) — *strictly smaller* than the Zygmund-type $B_{\infty, \infty}^2$ that an earlier draft named (second-order characterizations, $(I - T_t)^2$, would be needed for the latter; the correction is a referee's and we record it) — and regularity beyond the saturation class is invisible to this detector.

The converse and its correction (main contribution). Does a rate-1 reading certify saturation-class membership? For the practically available detector — the coefficient-free ratio form of Section 5, since the oracle form below needs the answer as input — over a finite window of scales, no: Section 3 gives the homogenization impostor — oscillatory microstructure c_δ reads, for $t \gg \delta^2$, as the *smooth* effective medium with the harmonic-mean coefficient (the classical 1d homogenization limit [5]), so that detector does not merely miss the fine structure, it performs the homogenization. We stress the logical type of this counterexample: c_δ is analytic, with finite (merely enormous) class norms, so the *asymptotic* ($t \downarrow 0$) converse is never violated — what fails is every fixed-window version of it, which is the version any finite measurement uses. The corrected converse must therefore be **scale-quantified**: full-rate decay at *all* scales (Conjecture C: in the semigroup-characterization frame [6], with the same undischarged proof obligation as Conjecture G). It is probed **constructively**: Section 5 demonstrates the inversion — class, position, amplitude — with blind classification at 12/12 including the impostor class, whose signature (a non-decaying or scale-broken reading within the window) is exactly what the graded law's exponent range $(0, 1]$ forbids for genuine Hölder classes at those scales.

Why this matters beyond spectral theory. If Conjectures G and C hold, the diagnostic upgrades to a characterization: "the coefficient is in the saturation class" and "the detector reads rate 1 at all scales" become interchangeable at stated scope, with the homogenized media

correctly *classified* (scale-break class) rather than misread. In applications where an effective geometry emerges from microstructure, by-scale classification is the physically meaningful statement.

2. Setting and the corrected detector

Unit circle, $L = -\partial_x(c\partial_x)$ with $0 < c \in L^\infty$, symmetric finite differences at $N = 512$, exact eigensolve; scale ladder $t \in [2.5 \times 10^{-4}, 6 \times 10^{-3}]$ (probing widths ≥ 8 grid cells).

Lemma 2.1 (lattice baseline). For constant c , the discrete kernel satisfies $K_t^{\text{disc}}(c) = K_{ct}^{\text{disc}}(1)$ exactly. Hence the lattice artifact admits the baseline $g(s) := r_s^{\text{disc}}(\cdot; c \equiv 1)$ and the **corrected detector** $\tilde{r}_t(x) = r_t(x) - g(c(x)t)$ removes it to first order in the variation of c .

Receipt: floor improvement from 1.5×10^{-2} to 6×10^{-7} ; the uncorrected pipeline's "measurements" are pure discretization (a documented failure of our own first run: smooth coefficients read rate -0.02 where the law says 1.0). We flag this prominently because any implementation of local-Weyl numerics faces it.

3. The finite-window converse is false: the homogenization impostor

Proposition 3.1 (impostor; classical limit plus receipts — stated for the right detector). Let $c_\delta(x) = 1 + a \sin(2\pi x/\delta)$. For $t \gg \delta^2$ the diagonal kernel is spatially flat and the **coefficient-free reading** — the fitted coefficient $\hat{c} = 1/(4\pi t K_t(x, x)^2)$, equivalently the ratio detector of Section 5 — reads c_δ as a smooth medium whose reported coefficient is the harmonic mean $\bar{c}_H = (\int c^{-1})^{-1}$; for $t \ll \delta^2$ the reading breaks away at $O(1)$.

Which detector is fooled — a correction kept visible. An earlier draft stated this for the oracle detector $\tilde{r}_t$ of Lemma 2.1. That was wrong, as a referee's reconstruction showed: $\tilde{r}_t$ normalizes by the *true* $\sqrt{c(x)}$, which oscillates at $O(1)$ against the homogenized (flat) kernel, so the oracle detector reads $\sup |\tilde{r}_t| \approx \sqrt{(1+a)/\bar{c}_H} - 1 \approx 0.41$, *non-decaying at all scales above the crossover* — it is never fooled, and that non-decaying signature is precisely how the blind classification of Section 5 catches the microstructure (fitted rate ≈ -0.07 there). The impostor phenomenon — and hence the finite-window failure of the converse — belongs to the *coefficient-free* detectors, which are also the only ones a real measurement has. The narrative of Sections 1 and 5 is stated for those.

Status of the statement. The $t \gg \delta^2$ regime is a consequence of classical periodic homogenization [5] (G-convergence of L_δ to the harmonic-mean operator, hence locally uniform convergence of the kernels); we do not re-prove it. What we add, and verify by machine rather than prove, is the *detector-level quantification*: the crossover location, the $O(1)$ breakaway size, and the fidelity of the reported coefficient at finite δ .

Receipts ($\delta = 1/32$, $a = 0.6$): fitted effective coefficient 0.8045 vs $\bar{c}_H = 0.8000$ (within 0.6% ; because the profile here is deterministic, the comparison carries no realization-versus-ensemble gap — unlike the random-coefficient example of Section 6, where that gap must be accounted for — and the arithmetic mean 1.0 is excluded at 25%); oracle detector above crossover: $\sup |\tilde{r}| \approx$

0.41, non-decaying, as the correction paragraph states; below crossover both readings break at $O(1)$ ($\sup |\tilde{r}| = 0.52$).

Note again the type of the example: c_δ is analytic. The impostor does not contradict the asymptotic converse; it shows that the converse *as restricted to any finite scale window* — the only version a finite-resolution measurement can instantiate — is false, and that the failure is physical (effective-medium formation), not pathological. The failure mode dictates the fix:

Conjecture C (corrected converse). If the coefficient-free detector reads rate $1 - O(t)$ decay — *at every scale* $t \downarrow 0$, then c belongs to the Favard/saturation class $\{c : c'' \in L^\infty\}$ of Section 1; more generally the all-scale rate $t^{\alpha/2}$ characterizes $B_{\infty,\infty}^\alpha$ membership for $\alpha \in (0, 1)$ (the Hölder range, where first-order semigroup characterizations [6] apply). The mechanism is the two-sided window estimate of Section 4; the proof obligation (uniform two-sided parametrix bounds for non-smooth coefficients) is open and stated, not papered over.

4. The two-sided mechanism

The kernel window bounds the local oscillation of c at scale $\sqrt{t}$ from both sides. The load-bearing receipt is the **amplitude sandwich**: for a $C^{0.5}$ cusp of amplitude η at fixed scale,

eta:	0.2	0.4	0.8	1.6
signal:	0.0146	0.0287	0.0556	0.1046
signal/eta in [0.0654, 0.0730] (two-sided, spread x1.12)				

A detector zero at scale t forces oscillation zero at scale $\sqrt{t}$; iterating over the ladder forces the modulus of continuity at every scale — Besov membership. Conversely the forward graded law bounds the signal above by the class norm. At the stated 1d scope this is the mechanism behind Conjecture C for $\alpha \in (0, 1)$ (the Hölder range; at $\alpha = 1$ the all-scale rate $t^{1/2}$ tracks the Zygmund class $B_{\infty,\infty}^1$, not Lipschitz, and we make no equivalence claim there) plus the saturation endpoint.

5. The constructive inversion

From detector data alone:

- **Class:** $\hat{\alpha} = 2 \times$ (fitted rate), capped at smooth. Across random positions and amplitudes ($\alpha \in \{0.5, 1.0\}$): $\max |\hat{\alpha} - \alpha| = 0.03$.
- **Position:** $\arg \max_x |\tilde{r}_t(x)|$ at the smallest valid scale lands within 7×10^{-4} of the cusp (kernel width 1.6×10^{-2}): the inversion is local.
- **Blind classification** over 12 cases (5 smooth random Fourier media, 5 cusps with random parameters, 2 microstructures): decision rule — fit residual > 0.25 *or* rate < 0.1 declares SCALE-BREAK; rate ≥ 0.85 declares SMOOTH; otherwise CUSP with $\hat{\alpha}$. Result: 12/12. Threshold honesty: the rate *floor* (0.1) is principled, not tuned — graded exponents live in $(0, 1]$, so a non-decaying reading (microstructure enters at fitted rate ≈ -0.07 with small residual — a power law, but an inadmissible one) belongs to no Hölder class — while the residual threshold (0.25) and the smooth threshold (0.85) *are* tuned choices, and the 12/12 should be read with that asymmetry in mind.

The three-way output — smooth / Hölder / scale-break — is the correct taxonomy: the third class is *physics* (effective media), not noise, and the constructive converse classifies it instead of being fooled by it.

De-circularizing the detector. A fair objection: $\tilde{r}_t$ contains $\sqrt{c(x)}$ and $g(c(x)t)$ — the inversion above consults the very coefficient it claims to probe. Two remarks. First, the naive repair (estimate $\hat{c}$ from K_t at the smallest window scale and substitute) *fails by construction*: the detector then vanishes at the calibration scale and the decay fit measures the calibration, not the roughness — we ran it, and the fitted exponents pin at the cap; we record this as a warning to implementers. The correct repair is the **ratio detector**

$$s_\lambda(x, t) = \frac{K_{\lambda t}(x, x) \sqrt{\lambda} (1 + g(\hat{c}(x)t))}{K_t(x, x) (1 + g(\hat{c}(x)\lambda t))} - 1,$$

in which the unknown $\sqrt{4\pi c(x)t}$ normalization cancels *exactly* between numerator and denominator; a crude $\hat{c}$ (smallest-scale estimate) survives only inside the lattice correction g , where its $O(t^{\alpha/2})$ error multiplies a term that is itself $\lesssim 10^{-3}$. The roughness signal survives with the same exponent: $\sup_x |s_\lambda| \sim t^{\alpha/2}(\lambda^{\alpha/2} - 1)$. Receipt ($\lambda = 4$, three random cusp positions/amplitudes per class, both grids):

alpha = 0.5:	fitted rates 0.224 - 0.237	(law: 0.25)
alpha = 1.0:	fitted rates 0.489 - 0.495	(law: 0.50)
smooth:	rate 0.944 / 0.945	(law: ~1)

The inversion needs no oracle access to c .

6. The adversarial wall and honest limitations

Adversarial study. Fix a genuine $C^{0.5}$ cusp of amplitude η and minimize the detector excess over smooth-plus-oscillatory dressing (nine parameters, local search):

smooth control excess:	0.072
eta = 0.2 / 0.4 / 0.8:	0.306 / 0.538 / 0.929 (minimized)

Monotone in η , never approaching the control: in this dressing class a cusp cannot be hidden from the ladder. *Limitations*: the dressing class is small and the search local — the wall is search-bounded; a certificate version (sum-of-squares over the dressing class) is the natural upgrade.

The $\alpha \in (1, 2)$ band. Our grid rounds the cusp at scale h , mixing the $t^{\alpha/2}$ signal with the smooth tail; measured 0.49 at $\alpha = 1.5$ versus predicted 0.75. The equivalence is claimed for $\alpha \leq 1$ plus the saturation endpoint; closing the band needs finer grids or higher-order discretizations and is stated as open, not papered over.

Grid refinement. The headline rates are not lattice artifacts: refining $N = 512 \rightarrow 768$ (a $1.5\times$ finer grid) leaves every fitted rate unchanged to three decimals — 0.232/0.494/0.969 vs 0.232/0.494/0.970 for $\alpha = 0.5/1.0/\text{smooth}$ — and the de-circularized rates of Section 5 move by < 0.013 .

Dimension: the 2d port, executed. The parametrix mechanism and the Besov frame are dimension-generic, and we verify the port on the 2-torus ($L = -\nabla \cdot (c\nabla)$, 40^2 grid, the same

exact rescaling baseline — valid in any dimension):

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smooth rate 0.880 (law: 1.0, window-limited); cusp rates
0.197 / 0.452 at alpha = 0.5 / 1.0 (law: 0.25 / 0.50);
position recovery to 0.015 (kernel width 0.035).
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THE 2d IMPOSTOR has its own exact classical anchor: for a
statistically isotropic symmetric two-phase medium the
effective coefficient is the GEOMETRIC MEAN (Dykhne/Keller
duality [11]). Random checkerboard (c1, c2) = (0.5, 2.0):
kernel-reported effective c = 1.0057 vs geometric mean 1.0000
(0.6%). Quantity bookkeeping, stated: 1.0000 is the ENSEMBLE
Dykhne/Keller target (symmetric fractions); the random
realization has phase fraction p-hat ~ 0.53, whose own
arithmetic mean is the quoted 1.205 (not the ensemble 1.25),
and realization fluctuation contributes at the same order as
the quoted 0.6% — so the receipt is "ensemble target matched
to within realization noise", with the arithmetic mean
excluded at 20%. Below the cell scale the reading breaks
at 0(1).
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The detector thus performs the *correct, dimension-specific* homogenization in 2d exactly as it performed the harmonic-mean homogenization in 1d — strong evidence that the corrected (all-scale) converse is the right statement in every dimension. The general- d theorem with anisotropic coefficient classes, and a Lorentzian port (Euclidean-first through Osterwalder–Schrader rotation), remain the named sequels.

7. Related work

Heat invariants and local Weyl expansions for smooth coefficients are classical [1, 2, 3]; regularity detection via semigroup approximation rates is implicit in the Besov characterization literature [6]; homogenization of oscillatory coefficients is textbook [5]. Two adjacent literatures must be distinguished from what is done here. *Inverse spectral theory* (Borg [7]; Gel’fand–Levitan [8]; Marchenko [9]) recovers the coefficient itself from full spectral data (spectrum plus norming constants); we use far less input — the decay *rate* of the on-diagonal kernel — and obtain a correspondingly coarser output, a regularity class with localization, which is why the method survives non-smooth coefficients where the classical machinery assumes regularity. *Aronson-type Gaussian bounds* [10] guarantee that K_t and hence the detector are well-defined for merely measurable uniformly elliptic c , which is what licenses evaluating $\tilde{r}_t$ on every class discussed, including the microstructures. The contributions here are (i) the graded *local sup* law with receipts at non-smooth classes, (ii) the impostor proposition identifying the precise failure (and physical meaning) of the finite-window converse, (iii) the constructive inversion with the principled three-way decision rule and its de-circularized variant, and (iv) the adversarial formulation of detector rigidity.

Reproducibility

Fixed-seed scripts regenerate every number: the baseline lemma and floor receipts; the graded rates; the homogenization receipts; the inversion and blind classification; the adversarial walls.

Code and canonical outputs accompany the submission.

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